2.4 COMPRESSION

The Compression Section include one Syn. Gas Compressor & Circulator (K-301), one Process Air Compressor (K-302), and one Natural Gas Compressor (K-303). The Syn. Gas Compressor & Circulator is a three-casings machine which compresses the make-up gas to 239 Kg/cm^2G, and Circulates the recycle synthesis gas through the synthesis loop. The Process Air Compressor is a three-casings machine which compresses atmosphere air to 34 Kg/cm^2G. The Natural Gas Compressor is a one-case machine which compresses natural gas to 40 Kg/cm^2G. All three compressors are the centrifugal type driven by steam turbines.

a. Synthesis Gas Compressor (K-301)

The Syn. Gas Compressor & Circulator consists of the four make-up stages and the recycle stage. Each make-up stage has the intercooler which cools the synthesis gas to 43°C for saving the power of compression and the suction drain separator which protects the compressor from the troubles of liquid suction. The first stage suction pressure 25 Kg/cm^2G is maintained by the control of turbine speed, the final discharge pressure of make-up stage is about 239 Kg/cm^2G at normal speed.

The compression head of Circulator is designed 28.6 Kg/cm^2G at normal speed.

This system has the safety device preventing the compressor from surging; anti-surge bypass control. Minimum suction flow rate not to surge is assured by recycling flow.

Synthesis gas from the Methanator passes through the cooling water cooled Synthesis Gas Precooler (E-301) and Synthesis Gas Precooler Separator (V-301) which provide a suction gas temperature of 43°C and a suction gas pressure of 25 Kg/cm^2G at the inlet of the L.P. casing. After the first compression step, synthesis gas passes through the First Stage Synthesis Gas Cooler (E-302) where heat of compression is removed. Any water condensed is removed in the First Stage Synthesis Gas Separator (V-302). The synthesis gas then flows into the remaining stage of the Compressor. After each stage, the heat of compression is removed in the Interstage Coolers and the condensed water is separated in Interstage Separators. In the high pressure case synthesis recycle gas, separated from the make-up gas by a diaphragm, is boosted to 239 Kg/cm^2G before being recirculated to the synthesis loop. The first stage suction pressure is maintained by control of the Turbine speed. Separate anti-surge controls are provided for the make-up and recycle gas streams. The flow required to avoid surge in the first and second stage stream is measured at the first stage suction, and is maintained by recycling make-up gas from the second stage discharge. The flow required to prevent surge in the third and fourth stage stream, measured at the third stage suction, is maintained by returning make-up gas from the fourth stage discharge. The flow required to prevent surge in the recycle stream, measured at the recycle stage suction, is maintained by returning recycle gas to the recycle stage suction through the Water-Cooled Condensers.